

Phase 1 — LightGBM Baseline Report (E1)

Dataset: IEEE-CIS Fraud Detection (Vesta Corporation / Kaggle Competition) **Type:** Real-world-derived, anonymized, historical e-commerce transactions

This is a baseline experiment. No GRU, Transformer, Kafka, or adaptive retraining. The primary metric is **PR-AUC** (Average Precision). Accuracy is not reported.

Dataset

- Training rows: 434,176
- Validation rows: 77,822
- Test rows: 78,542
- Training fraud: 15,252 (3.513%)
- Features: 406

Temporal Split

Split	Days	Rows	Fraud %
Train	1–127	434,176	3.513%
Validation	127–155	77,822	3.389%
Test	155–183	78,542	3.532%

Preprocessing

- JOIN: LEFT JOIN train_transaction + train_identity on TransactionID
- Temporal features: hour_sin, hour_cos, day_index (from TransactionDT)
- Missingness indicators: binary flags for key high-missing columns
- Numerical imputation: median (fit on train only)
- Categorical encoding: LabelEncoder (fit on train only)
- Dropped: D6,D7,D8,D9,D12,D13,D14 (>80% missing), dist2 (93.6% missing)
- Dropped: V-columns with >80% missing in training data
- Excluded: TransactionID, isFraud, TransactionDT (raw)

Model Configuration

- Model: LightGBM 4.7.0
- Objective: binary
- Metric: average_precision (PR-AUC)
- is_unbalance: True (class imbalance handling)
- Best iteration: 1000
- Training time: 50.66s

Validation Results (threshold frozen here)

Metric	Value
PR-AUC	0.584935
ROC-AUC	0.923115
Precision	0.628479
Recall	0.513841
F1	0.565408
MCC	0.554723
Balanced Accuracy	0.751594
FPR	0.010654
FNR	0.486159
Decision threshold	0.616521

Final Test Results (evaluated ONCE with frozen threshold)

Metric	Value
PR-AUC	0.531731
ROC-AUC	0.898993
Precision	0.581810
Recall	0.493511
F1	0.534035
MCC	0.520301
Balanced Accuracy	0.740262
FPR	0.012987
FNR	0.506489

TP=1369 FP=984 TN=74784 FN=1405

Top Features (by Gain)

Rank	Feature	Gain
1	V258	792886.35
2	C13	454330.72
3	V294	393786.40
4	card1	363974.53

Rank	Feature	Gain
5	TransactionAmt	345534.19
6	day_index	314009.21
7	card2	292409.57
8	addr1	245972.37
9	C8	194804.98
10	D15	174443.87
11	C1	173555.04
12	M4	172864.70
13	D2	171575.03
14	card6	145283.47
15	V70	144995.73

Limitations

1. This is a baseline — no hyperparameter tuning (Optuna) performed.
2. No GRU temporal sequence modeling (Phase 2).
3. No concept drift adaptation (Phase 3).
4. No streaming evaluation (Phase 4).
5. SHAP computed on validation sample only.
6. Dataset is anonymized — V-column interpretations are not available.
7. IEEE-CIS is e-commerce fraud, not UPI transaction fraud.

Reproducibility

- Random seed: 42
- Python: 3.13.13
- LightGBM: 4.7.0
- pandas: 3.0.5
- numpy: 2.4.6

Reproduction command:

```
python experiments/run_phase1.py --config configs/phase1_lightgbm.yaml
```

Next Recommended Phase

Phase 2: Temporal sequence modeling using GRU on card1 entity histories. Requires: Phase 1 model as tabular baseline for comparison.